ThermogramBaseline Demo

In order to do some of the work below, we must have the tidyverse package installed.

```
library(tidyverse, quietly = TRUE)
#> -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
#> v dplyr     1.1.4     v purrr     1.0.2
#> v forcats   1.0.0     v readr     2.1.5
#> v ggplot2   4.0.1     v tibble    3.2.1
#> v lubridate 1.9.3     v tidyr     1.3.2
#> -- Conflicts ------------------------------------------ tidyverse_conflicts() --
#> x dplyr::filter() masks stats::filter()
#> x dplyr::lag()    masks stats::lag()
#> i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

Here we load the package and look at the data. Data must be formatted in this way:

```
library(ThermogramBaseline)
```

```
data(DemoBaseline, package = "ThermogramBaseline")
head(DemoBaseline)
```

SampleID	Temperature	dCp
3a	20.00018	0.0899742
3a	20.03254	-0.0816733
3a	20.08539	-0.0622706
3a	20.15371	0.0106867
3a	20.25106	0.1488339
3a	20.34125	0.1822992

Each function can only take in one thermogram sample at a time so we must filter for sample 11b and filter the temperature to between 45 and 90

```
SampleIDs <- DemoBaseline %>% pull(SampleID) %>% unique() %>% as.vector()

Sample.1 <- DemoBaseline %>% filter(SampleID == SampleIDs[2]) %>%
  filter(between(Temperature, 45,90))
```

Now that we have our first sample, lets get our two endpoints for baseline detection.

```
endpoints <- endpoint.detection(x = Sample.1, exclusion.lwr = 60,
                                exclusion.upr = 80, point.selection = "innermost",
                                explicit = FALSE)
```

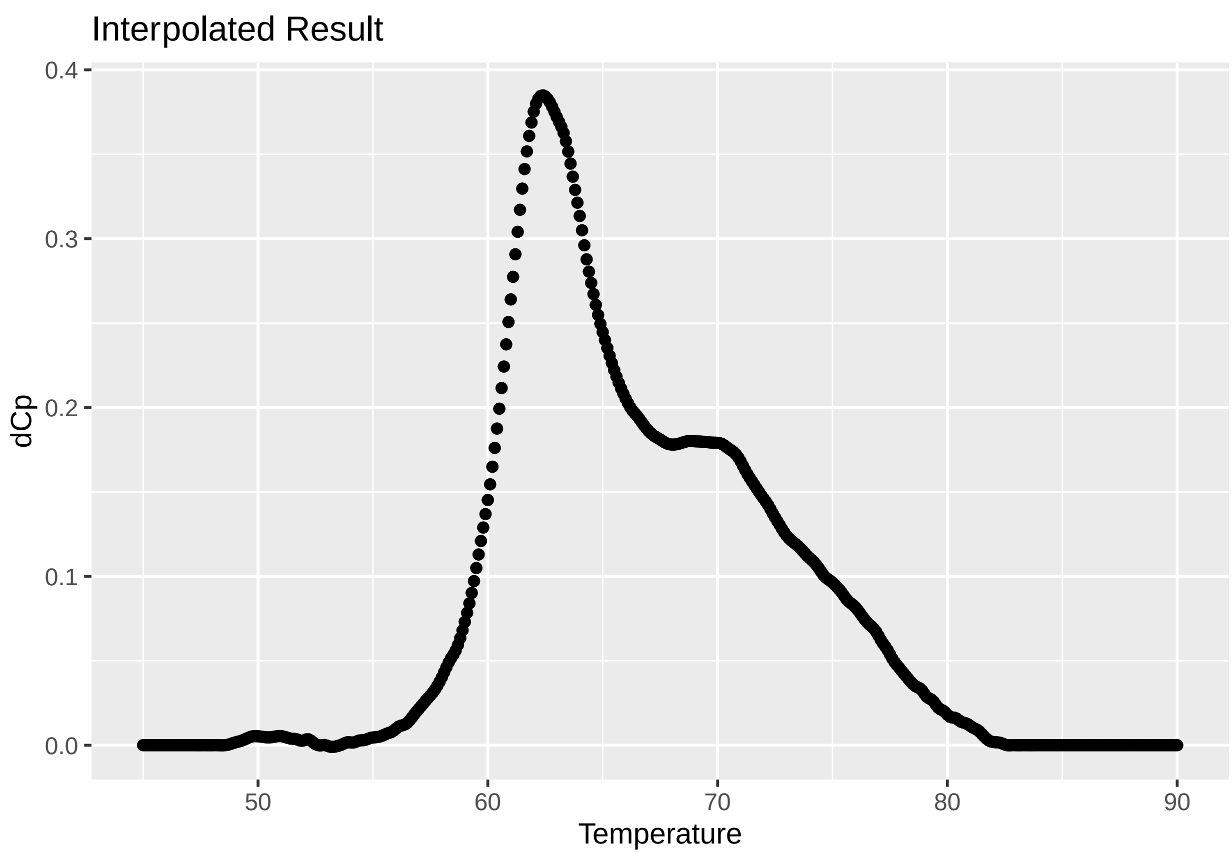
lower	upper	method
48.63918	82.55743	innermost

The model selected the temperatures 48.63918 and 82.55743 as our endpoints for our baseline subtraction. Next lets subtract the baseline using these endpoints.

```
baseline <- baseline.subtraction.byhand(x= Sample.1, lwr.temp = endpoints$lower,
                                        upr.temp = endpoints$upper, plot.on = FALSE)
```

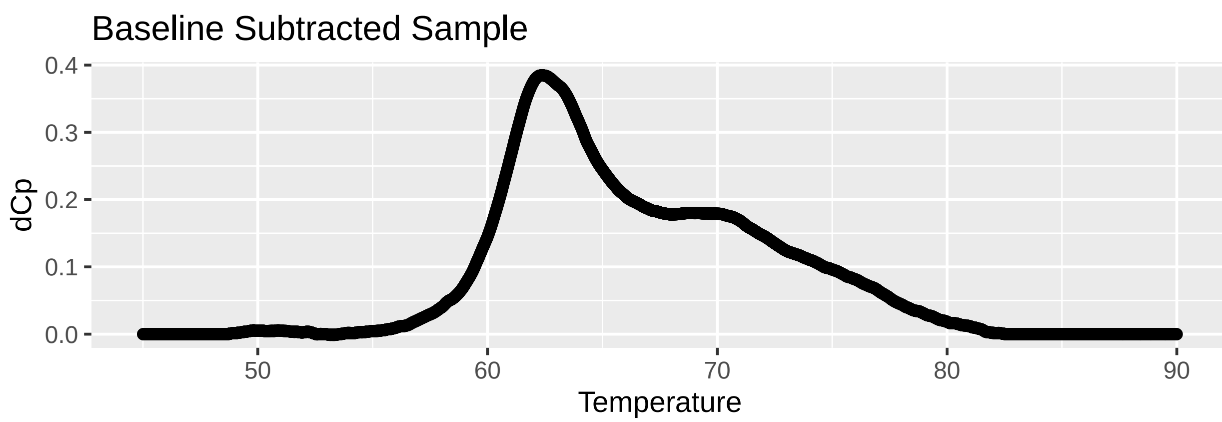
Now that we have subtracted the baseline, we will interpolate the data onto a grid of set temperatures. Notice plot.on is true by default.

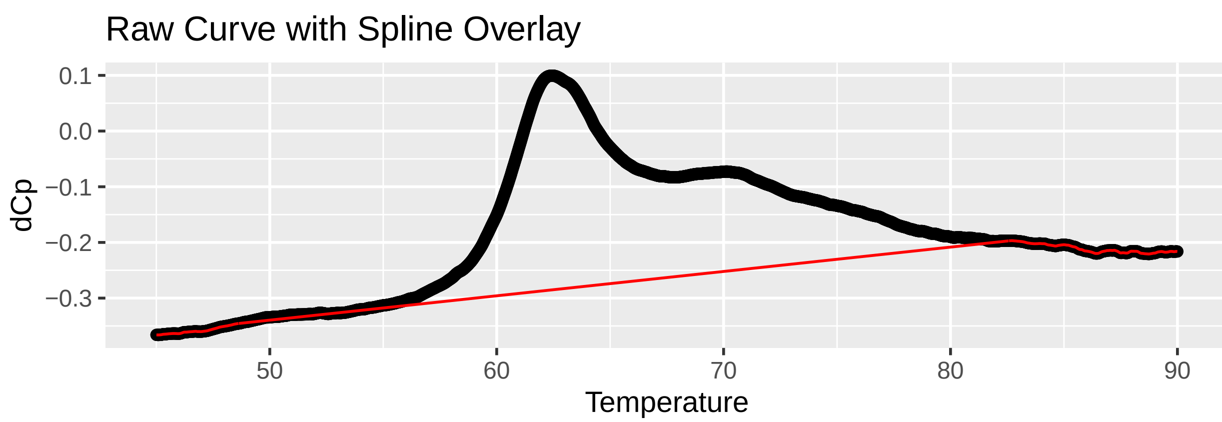
```
final <- final.sample.interpolate(x=baseline,grid.temp = seq(from = 45,to = 90,by = 0.1))
```

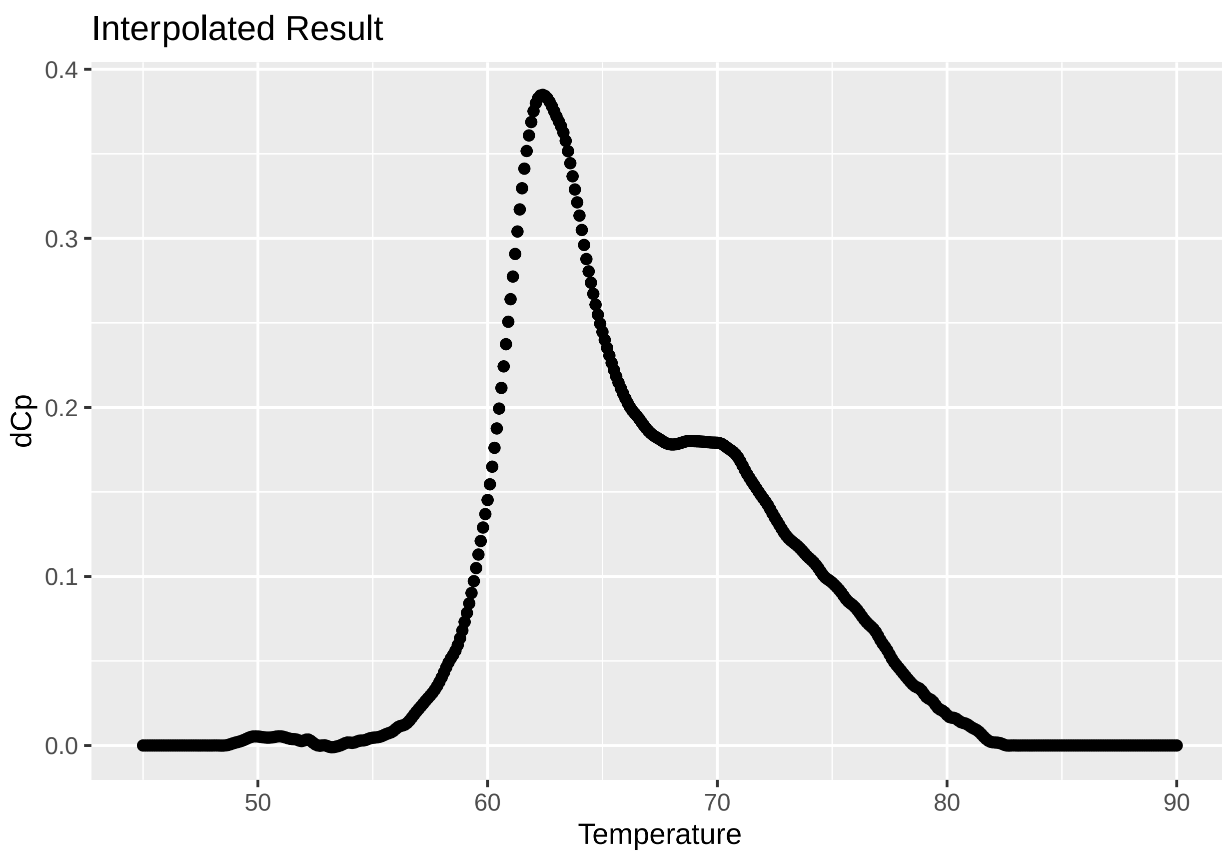


Here we have our final baseline subtracted interpolated sample. We can also use the auto function to complete these in one step. Notice that the plot.on default is false so we must set it to true if we want graphs to appear.

```
auto <- auto.baseline(x=Sample.1, exclusion.lwr = 60, exclusion.upr = 80,
                      grid.temp = seq(45,90,0.1),plot.on = TRUE, explicit = FALSE)
```


Baseline Subtracted Sample
dCp
0.4
0.3
0.2
0.1
0.0
50
60
70
80
90
Temperature


Raw Curve with Spline Overlay
dCp
0.1
0.0
−0.1
−0.2
−0.3
50
60
70
80
90
Temperature



Lastly, multiple.auto.baseline() will allow for multiple thermograms to be input at once. This will return one dataframe with all thermograms cleaned and returned on the same scale.

```
all_clean_thermograms <- multiple.auto.baseline(DemoBaseline)
#> Working on Sample  3a element  1  of 6
#> Scanning Lower.
#> Scanning Upper.
#> Working on Sample  11b element  2  of 6
#> Scanning Lower.
#> Scanning Upper.
#> Working on Sample  12a element  3  of 6
#> Scanning Lower.
#> Scanning Upper.
#> Working on Sample  15b element  4  of 6
#> Scanning Lower.
#> Scanning Upper.
#> Working on Sample  16b element  5  of 6
#> Scanning Lower.
#> Scanning Upper.
#> Working on Sample  29a element  6  of 6
#> Scanning Lower.
#> Scanning Upper.
#>
```

```
kable(head(all_clean_thermograms))
```

Temperature	3a	11b	12a	15b	16b	29a
45.0	-3e-07	-0.0001197	2.3e-06	0.0052488	-0.0023416	-0.0003207
45.1	-4e-07	-0.0001320	2.6e-06	0.0054906	-0.0022340	-0.0003168
45.2	-4e-07	-0.0001443	2.8e-06	0.0057360	-0.0021124	-0.0003080
45.3	-5e-07	-0.0001564	2.8e-06	0.0059848	-0.0019812	-0.0002937
45.4	-5e-07	-0.0001683	2.7e-06	0.0062367	-0.0018448	-0.0002731
45.5	-5e-07	-0.0001794	2.4e-06	0.0064916	-0.0017076	-0.0002451